Announcements

Quiz 5: this Wed.

Midterm 3 next Wed. (4/29) in class

Policy email coming soon

Midterm 2 grades released + sol'ns posted

Mean: 64.2/78

Median: 64/78

Std. dev.: 7.7

Q1: 69%

Q2: 81%

Q3: 94%

Q4: 100%

Q5: 90%

Gradelines:

A/A-: 66-78

B+/B/B-: 54-65.9

C+/C/C-: 40-53.9

D+/D/D-: 18-39.9

out of 78

Gradeline calculator updated

Regrade requests open for 1 week

Def: Let G be a graph w/ vertices $v_1, \dots, v_n$.
and edges $e_1, \dots, e_m$

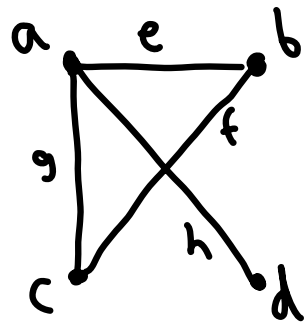
The incidence matrix of G is the matrix $\mathrm{Inc}_G = [m_{ij}]$

or both endpoints!

where $m_{ij} = \begin{cases} 1, \text{ if } v_i \text{ is an endpoint of } e_j \\ 0, \text{ otherwise} \end{cases}$

Ex:

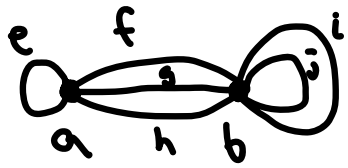
G:



$$\mathrm{Inc}_G = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} e & f & g & h \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Ex:

G:



$$\mathrm{Inc}_G = \begin{array}{c} \\ a \\ b \end{array} \begin{array}{c} \begin{array}{cccccc} e & f & g & h & i & j \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

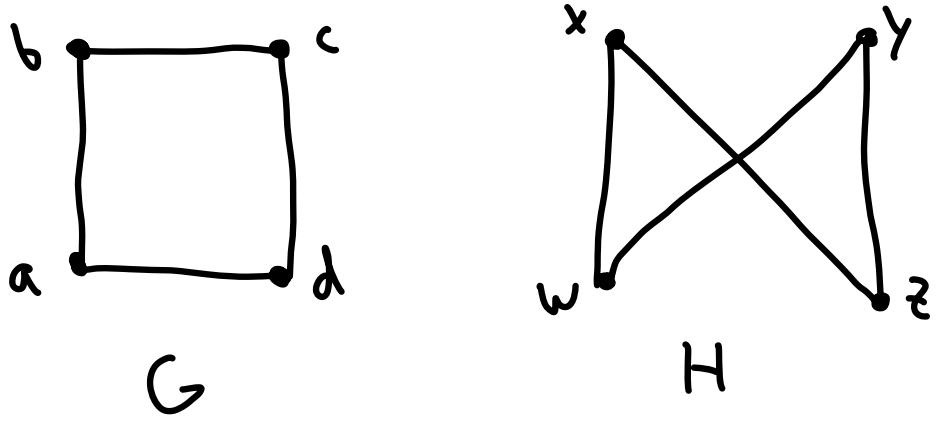
Def: Let $G = (V_1, E_1)$ and $H = (V_2, E_2)$ be simple graphs. A function $f: V_1 \to V_2$ is an <u>isomorphism</u> if

a) f is a bijection

b) $f(a)$ and $f(b)$ are adj. if and only if a and b are adj.

If any isomorphism exists, G and H are <u>isomorphic</u>

Ex 8:



Graph isomorphism:

$f(a) = w$ $\quad$ $f(c) = z$

$f(b) = x$ $\quad$ $f(d) = y$

Isomorphic graphs have to have the same:

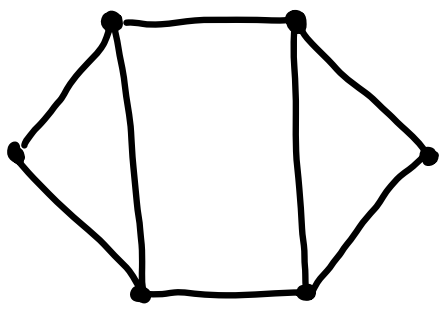
a) number of vertices

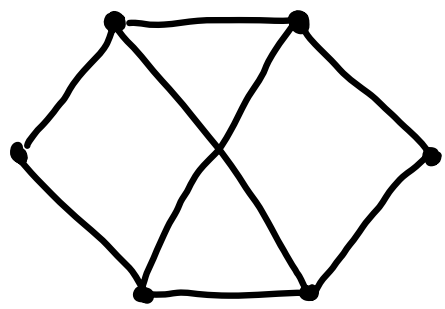
b) number of edges

c) lists of degrees

So if G & H differ on any of these $\rightsquigarrow$ not isomorphic!

Be careful:





same a), b), c), but not isomorphic!

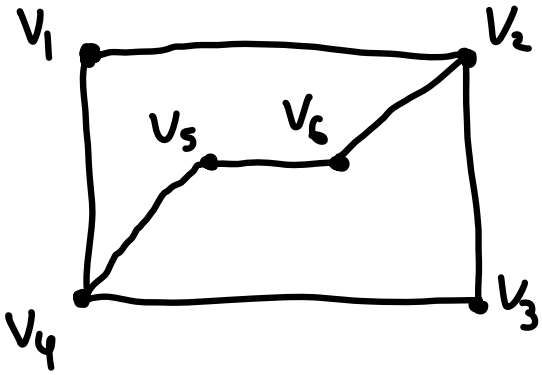
Two ways to show two graphs <u>are</u> isom.

1) Find an isomorphism

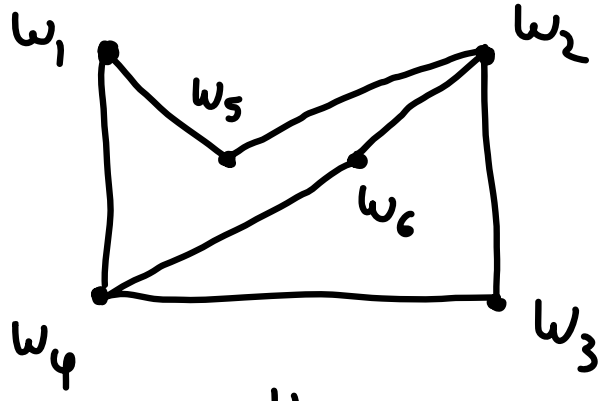
2) Show that the adjacency matrices are the same for some ordering of the vertices

(always same ordering on rows & cols!)

Ex 11:



G



H

$$\text{Adj}_G = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$\text{Adj}_H = \begin{array}{c} \\ w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{array} \begin{array}{c} \begin{array}{cccccc} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Not the same

But... put the vertices in a different order, and

$$
\begin{array}{c|cccccc}
 & w_6 & w_2 & w_3 & w_4 & w_1 & w_5 \\
\hline
w_6 & 0 & 1 & 0 & 1 & 0 & 0 \\
w_2 & 1 & 0 & 1 & 0 & 0 & 1 \\
w_3 & 0 & 1 & 0 & 1 & 0 & 0 \\
w_4 & 1 & 0 & 1 & 0 & 0 & 0 \\
w_1 & 0 & 0 & 0 & 1 & 0 & 1 \\
w_5 & 0 & 1 & 0 & 0 & 1 & 0
\end{array}
$$

So G and H are isomorphic.

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§10.4: Connectivity

Def: A <u>path</u> is an alternating sequence

$$v_0, e_1, v_1, e_2, \ldots, e_n, v_n$$

$v_i \in V$, $e_i \in E$ $\quad e_i$ has endpoints v_{i-1} & v_i

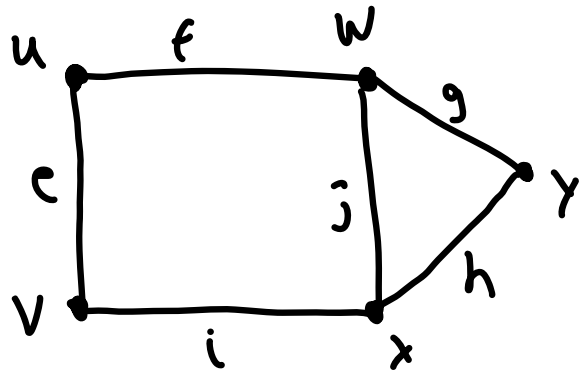
If no repeated edges, the path is <u>simple</u>

If $v_0 = v_n$, it is a <u>circuit</u>

If G is a simple graph, don't need to write the edges, since they're implies

* Note: terminology matches Rosen, different from other sources

Class activity:



Which of the following are paths? Circuits? Simple paths/circuits?

a) u, f, w, j, x, i, v, e, u

b) u, w, x, v, u

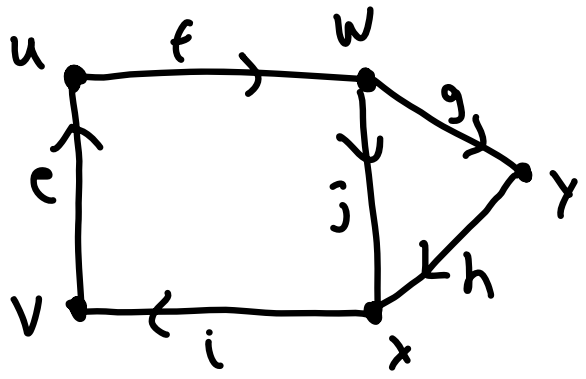
c) u, e, w, i, x, j, v, f, u

d) u, w, x, y, w, x, y

e) u, f, w, f, u, f, w

Def: Paths/circuits are the same in digraphs except they must <u>follow the arrow</u>

Class activity:



Which of the following are paths? Circuits? Simple paths/circuits?

a) u, f, w, j, x, i, v, e, u

b) u, w, x, v, u

c) u, e, w, i, x, j, v, f, u

d) u, w, x, y, w, x, y

e) u, f, w, f, u, f, w

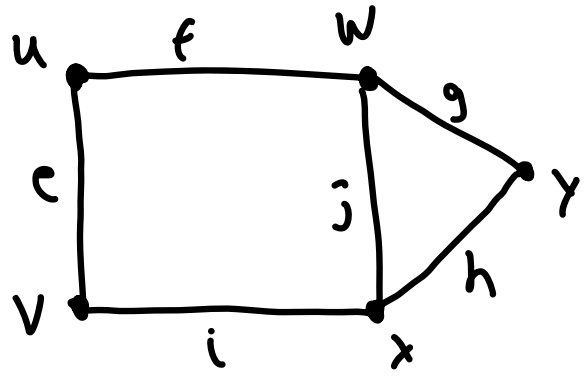
Def: a) A graph is <u>connected</u> if there is a path from every vertex to every other vertex

b) A digraph is <u>strongly connected</u> if there is a path from every vertex to every other vertex

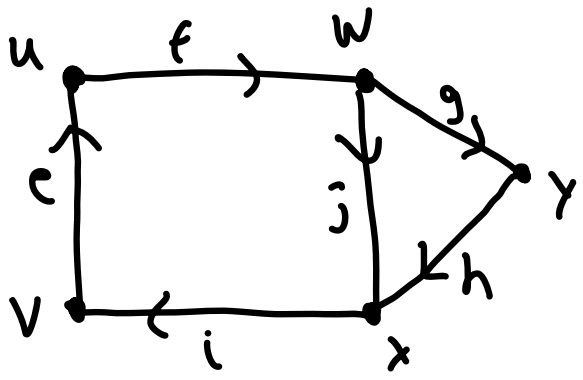
c) A digraph is <u>weakly connected</u> if there is a path from every vertex to every other vertex in the underlying graph (erase the arrows)

d) A <u>cut edge</u>/<u>cut vertex</u> in a graph is an edge/vertex which, when deleted, disconnects the graph

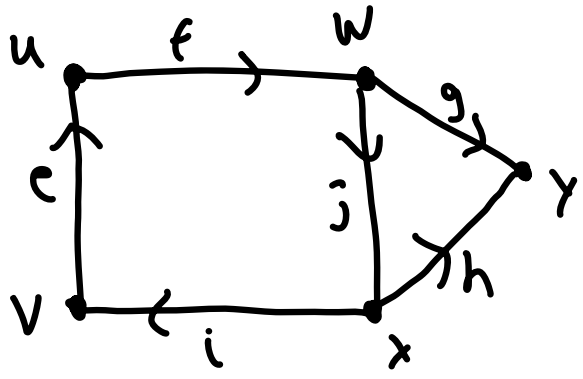
Examples:



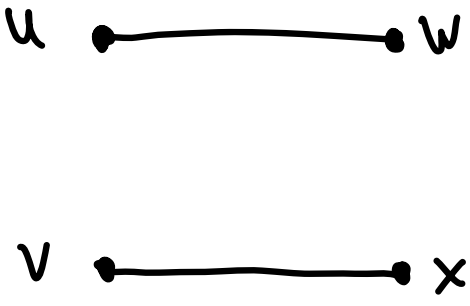
Connected, no cut-edges/vertices



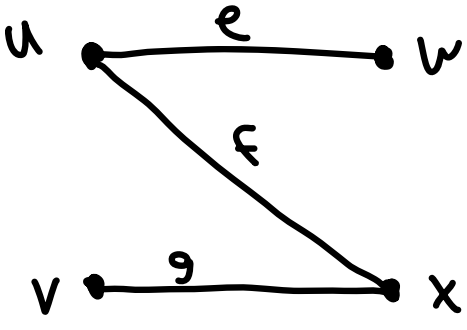
strongly conn.



weakly conn. since no path from y to e.g. v



disconnected



connected

cut-vertices : u, x

cut-edges : e, f, g